1D CNN — Architecture Variant Comparison

Generated 2026-05-17 21:02:09

Configuration (full)

Every parameter from the configuration cell is reproduced verbatim below. Copy the values back into the notebook config cell to regenerate this run. Derived counts (trials / channels / classes) are shown in italics in the Derived quantities section.

Data locations & class filter

DATA_PATHS[0]	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic
DATA_ROOT (legacy)	None
BATCH_TEST_ROOT	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
TEST_USER	8_Jestin
INCLUDE_LABELS	None
ReportPostfix	dense128

Sensor selection

Selection mode	SENSOR_GRID dict (fine-grained per-sensor)
USE_LEFT_HAND	True
USE_RIGHT_HAND	True
INCLUDE_SEGMENTS	['thumb', 'index', 'middle', 'ring', 'pinky', 'wrist']
SENSORS enabled (n)	44
SENSORS disabled (n)	0

SENSOR_GRID (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✓	✓	✓	✓
left index	✓	✓	✓	✓
left middle	✓	✓	✓	✓
left ring	✓	✓	✓	✓
left pinky	✓	✓	✓	✓
left wrist	✓	—	—	—
right palm	✓	—	—	—
right thumb	✓	✓	✓	✓
right index	✓	✓	✓	✓
right middle	✓	✓	✓	✓
right ring	✓	✓	✓	✓
right pinky	✓	✓	✓	✓
right wrist	✓	—	—	—

IMU channel modalities

USE_YPR	True
USE_QUAT	False
USE_ACCEL	True
USE_FLEX	False

Preprocessing

RESAMPLE_TO_N_STEPS	90
APPLY_BUTTERWORTH	True
BUTTERWORTH_CUTOFF_HZ	10.0
BUTTERWORTH_ORDER	4
SAMPLING_RATE_HZ	30.0
NORMALISATION	per_trial_zscore

Train / test split

TEST_SIZE	0.1
RANDOM_STATE	42
STRATIFY_SPLIT	True

Cross-validation

RUN_CROSS_VALIDATION	True
CV_FOLDS	7
CV_SHUFFLE	True

Augmentation (master switches)

AUGMENT_TRAINING_DATA	True

AUGMENTATION_COPIES_PER_SAMPLE	6
AUGMENTATION_RANDOM_SEED	42

AUGMENT_CONFIG (per transform)

Transform	Enabled	apply_prob	Other params
time_shift	✓	1.0	max_shift_steps=3, fill_mode=edge
time_warp	✓	1.0	speed_range=(0.95, 1.05)
time_mask	✓	1.0	max_masks=2, max_mask_size=3, fill_mode=zero
gaussian_noise	✗	1.0	std_ratio=0.02
amplitude_scale	✓	1.0	scale_range=(0.95, 1.05)
baseline_offset	✓	1.0	offset_std_ratio=0.02
channel_dropout	✓	1.0	drop_fraction=0.03, fill_mode=zero
left_right_swap	✗	0.5	—

Training hyperparameters

TRAIN_EPOCHS	60
TRAIN_BATCH_SIZE	16
TRAIN_VALIDATION_SPLIT	0.2
TRAIN_VERBOSE	1

Early stopping

EARLY_STOPPING_ENABLED	True
EARLY_STOPPING_MONITOR	val_loss
EARLY_STOPPING_PATIENCE	8
EARLY_STOPPING_MIN_DELTA	0.0
EARLY_STOPPING_RESTORE_BEST	True
EARLY_STOPPING_VERBOSE	1

Output paths

MODEL_OUTPUT_DIR	/home/jestin/ThesisRepo/ML/NewReports/NewDynamicReports/NN_Models
EXPERIMENT_LOG_PATH	/home/jestin/ThesisRepo/ML/NewReports/NewDynamicReports/NN_Models/experiment_results.csv

Variants in this run

Name	Builder	Description
Shallow	build_shallow	Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax
WideKernel	build_wide_kernel	Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

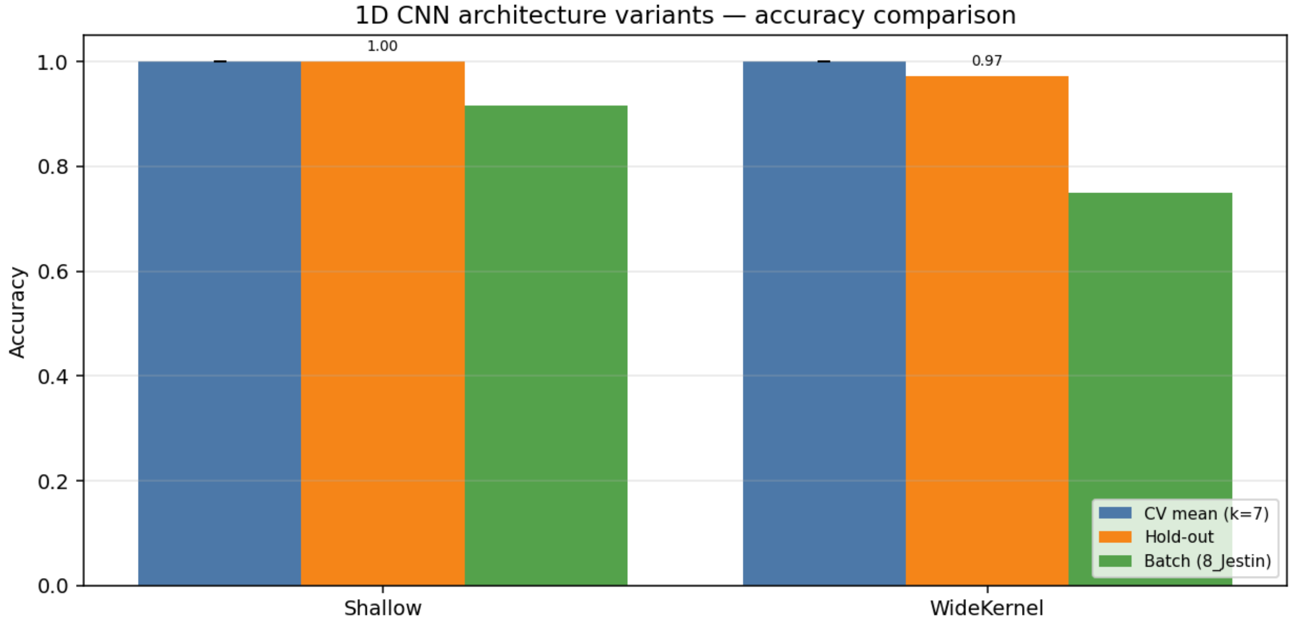
Derived quantities (read-only — computed from data)

Total trials (X.shape[0])	360
Train pool size	324
Hold-out size	36
Sequence length	90
Sensor channels	144
Number of classes	6
Class names	Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap
len(SENSOR_COLS)	144

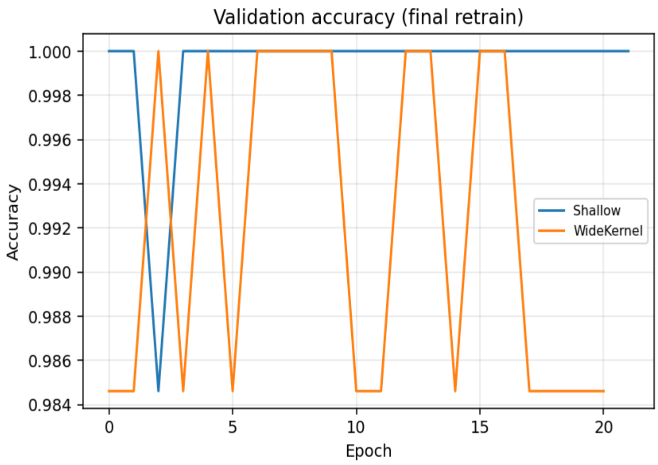
Variant comparison (summary)

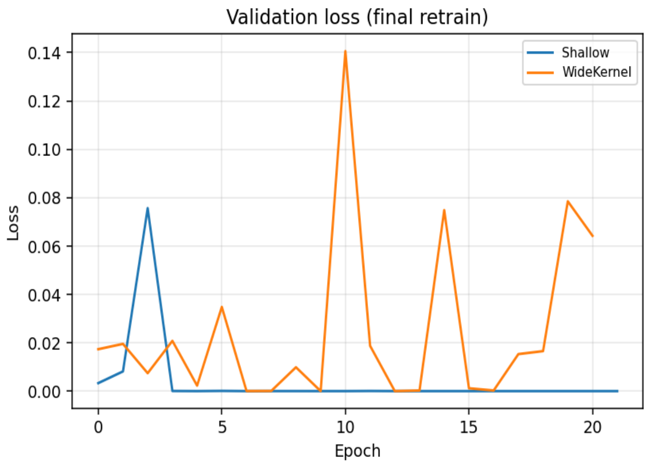
Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Shallow	67,334	1.0000	0.0000	1.0000	0.0001	0.9167	33/36
WideKernel	123,430	1.0000	0.0000	0.9722	0.0269	0.7500	27/36

Best on hold-out: **Shallow** · Best on CV: **Shallow**



Training curves (final retrain on full training pool)





Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	67,334
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0001
Batch-test acc	0.9167 (33/36)

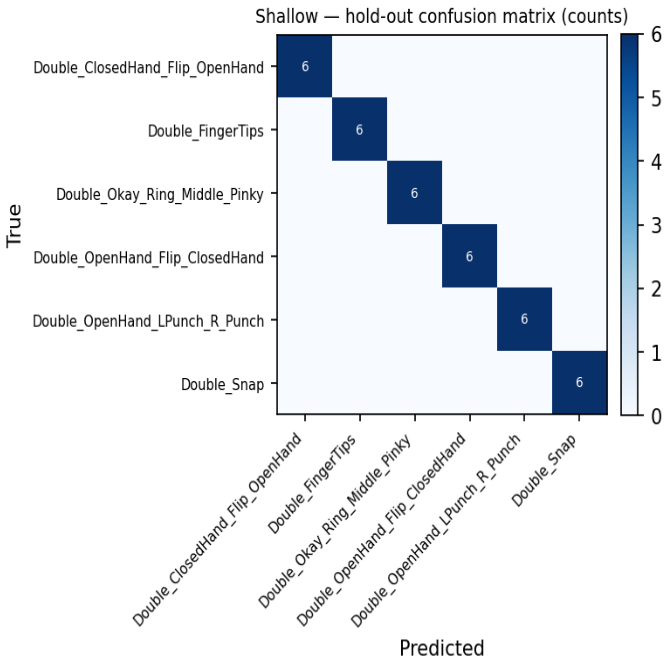
Cross-validation folds

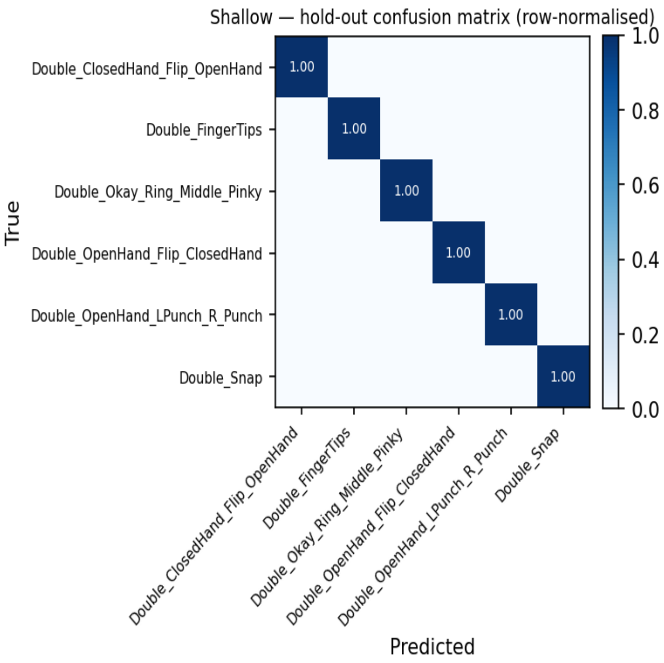
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

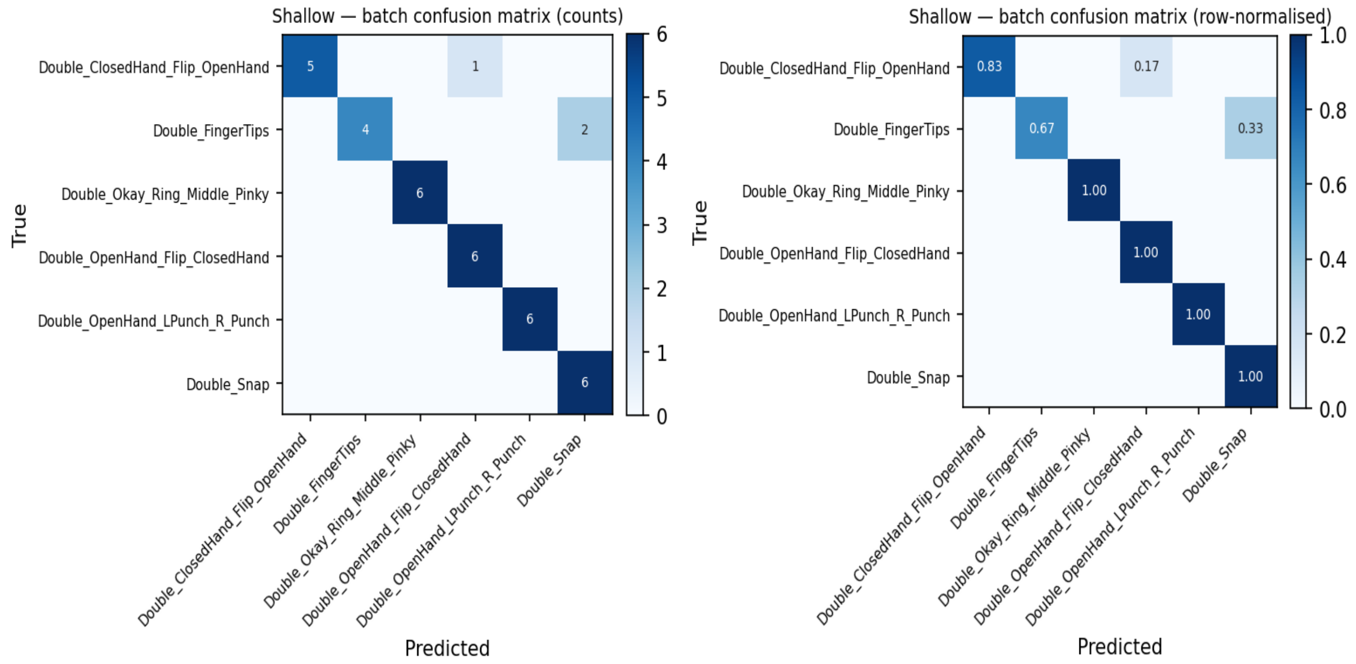




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	4	6	0.667
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	6	6	1.000
Double_Snap	6	6	1.000
Overall	**33**	**36**	**0.917**

Batch confusion matrix (8_Jestin)


Shallow — batch confusion matrix (counts)
Shallow — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	123,430
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	0.9722 · 0.0269
Batch-test acc	0.7500 (27/36)

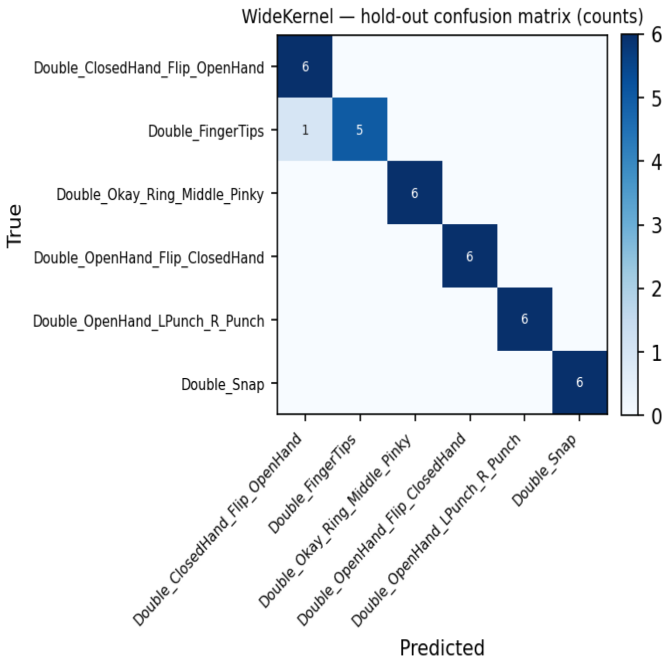
Cross-validation folds

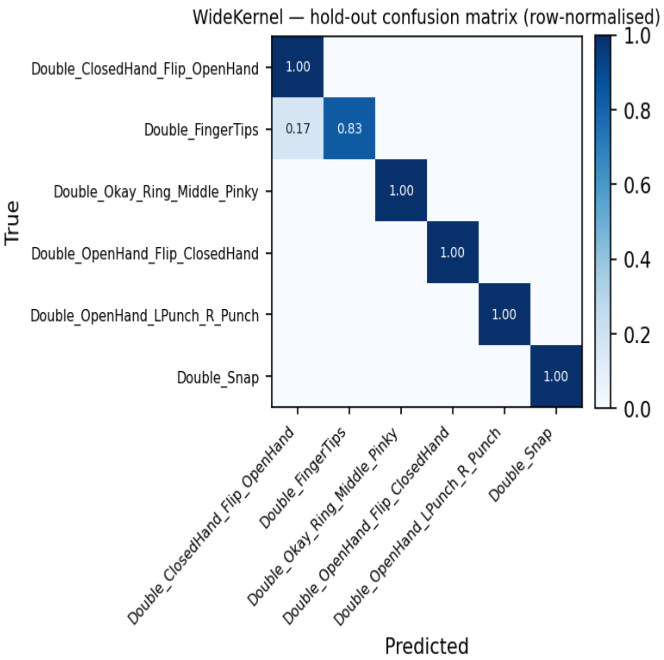
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	0.857	1.000	0.923	6
Double_FingerTips	1.000	0.833	0.909	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*0.976*	*0.972*	*0.972*	*36*
weighted avg	*0.976*	*0.972*	*0.972*	*36*

Hold-out confusion matrix

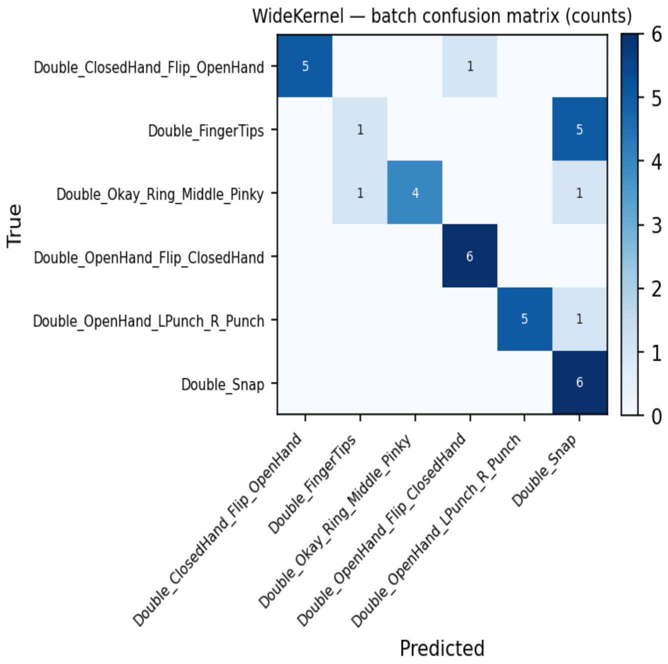


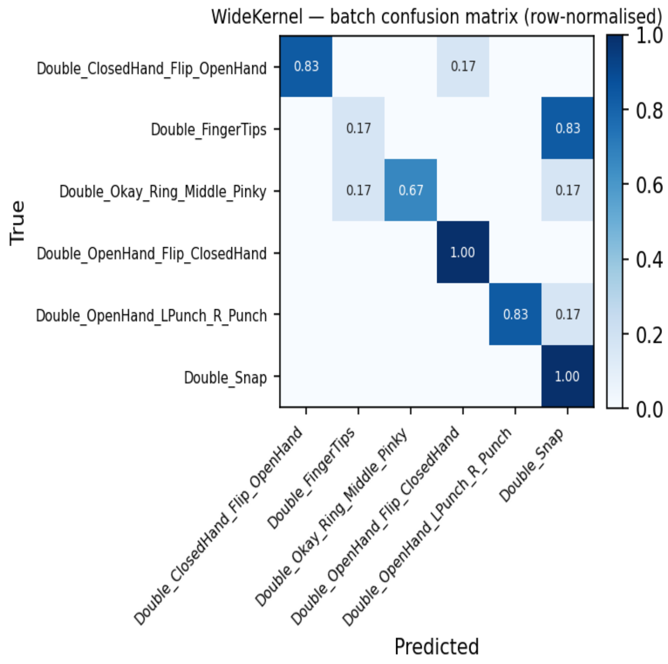


Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	1	6	0.167
Double_Okay_Ring_Middle_Pinky	4	6	0.667
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**27**	**36**	**0.750**

Batch confusion matrix (8_Jestin)


WideKernel — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap


WideKernel — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap